

Gradient Boosting (for Classification)

	Chest pain	Good Blood Circulation	Blocked Arteries	Heart disease	Residual
S ₁	No	No	No	No 0	0-0.66
S ₂	Yes	Yes	Yes	Yes 1	1-0.66
S ₃	Yes	Yes	No	Yes 1	1-0.66
S ₄	Yes	No	No	Yes 1	1-0.66
S ₅	Yes	No	Yes	Yes 1	1-0.66
S ₆	No	Yes	No	No 0	0-0.66

1) log odds

2) Convert to prob

Step 1: Initial prediction $\log(\text{odds}) \rightarrow -\infty \text{ to } +\infty$

Odds = $\frac{\text{Success}}{\text{no. of times failure}}$

$$\log(\text{odds}) = \log\left(\frac{4}{2}\right) \approx 0.7$$

Step 2: Convert to probability

$$p = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}} = \frac{e^{\log(4/2)}}{1 + e^{\log(4/2)}} = 0.66$$

Good Blood Circulation

Blocked
ArteriesChest
pain

0.3

0.3, -0.6

0.3, 0.3

-0.6

* affine - ai /
gradient boosting
trees for
classification.

 $\sum \text{Residual}_i$

3. Predict residuals & Convert the leaf values to log(Odds)

 $\sum \text{Residual}_i$ $\sum [\text{Previous Prob}_i \times (1 - \text{Prev Prob}_i)]$

$$\text{for 1st leaf} = \frac{0.3}{0.66 \times (1 - 0.66)}$$

$$= 1.43$$

$$\text{for 2nd leaf} = \frac{0.3 + (-0.6)}{0.66 \times (1 - 0.66) + 0.66(1 - 0.66)}$$

$$= \frac{-0.3}{0.2244 + 0.2244} = \frac{-0.3}{0.4488}$$

$$= -0.89$$

$$3^{\text{rd}} \text{ leaf} = 1.42$$

$$4^{\text{th}} \text{ leaf} = -3.3$$

Good Blood Circulation

Blocked arteries

Chest pain

1.43

-0.89

1.42

-3.33

log odd value

Repeat for further iterations:

convert to probability space

that value - prev residual (-0.66)